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Wind and Solar Radiation Complementarity Analysis

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Abstract

to be completed

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Introduction

This work seeks to explore under which geographical and temporal conditions wind speeds and solar radiation present complemenarity profiles in Switzerland.

I collect wind and solar radiation data from MeteoSwiss at hourly, daily and monthly level including measures of seasons, location and elevation. These data are at the origin of both wind and solar electricity production. In another work, Cauz et al. [2025] explore complementarity between wind turbines and solar panels (PVs) electricity production. Overall, they find a negative correlation between the two variables, mitigating risks to which renewables energy sources are exposed to.

I find that Wind–solar complementarity in Switzerland depends on the time horizon, season, and localisation. Negative correlation is higher in subsample relying on electricity production thresholds. Overall wind–solar complementarity is present across the country, but is stronger in the northern part of the country and when the elevation reaches at least 1200 meters.

1 Data and sample construction

I use data from MeteoSwiss from 2020 to 2023[2], relying on measures of mean wind speeds (in m/s) and solar radiations (in W/m^2) across Switzerland at hourly, daily and monthly level. The choice variable choice constitute an hypothesis of this work, I defend it in another document (changer le lien de direction). In short, electricity production from wind turbine is due to wind speed and wind speed depends on the area, obstacles, mountain corridor, temperature, *etc.* While solar radiation is the source of electricity production for PV.

I use data from 129 stations out of 159, since 30 of them have been lost because of a lack of values during the cleaning data process.

Stations are grouped into four elevation classes: <400 m, 400–<800 m, 800–<1,200 m, and ≥1,200 m. Geographic zones (North, South, East, and West) are defined around the median station coordinates, assigning each station according to its dominant north–south or east–west displacement. These groups are analytical subdivisions, not official climatic regions.

Hereafter I display a map showing stations position across Switzerland separated in four parts: North, South, East and West.

[2] I did this to get along with Cauz et al. [2025] work. Thus it gives me a sort of benchmark

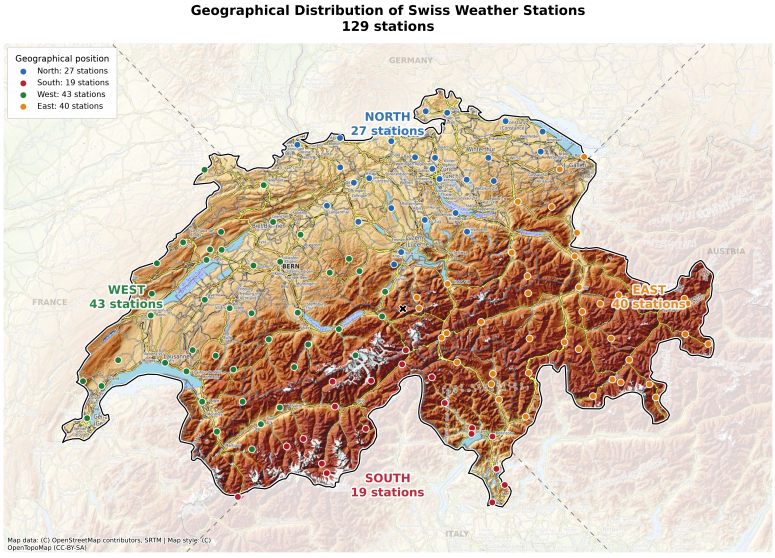

Figure 1: Geographical distribution of Swiss Weather Stations

An important point concerning wind and solar energy concerns threshold of production. When wind speed or solar radiation are too low, neither of technologies can produce electricity. Related literature report that, on average wind turbine begin to produce electricity from a wind speed about 3 m/s and cuts-out when wind speeds reach 25 m/s. While PVs start producing electricity at 100 W/m^2. On average, across all hours from 2020 to 2023, 29.4% of winds have the potential to make a turbine start while 61.9% of solar radiations have the potential to be at the origin of electricity production by PVs.

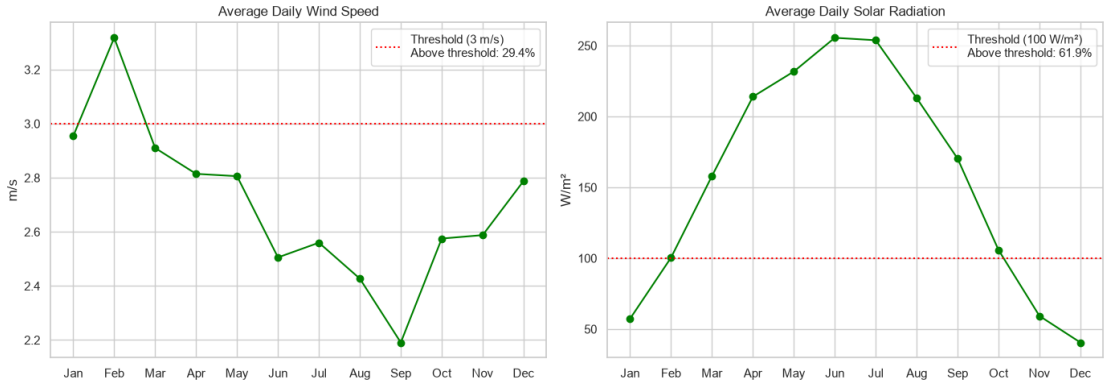

Figure 2: Proportion of measures above the given thresholds

re run la figure au cas où ça change dans le code.

Hereafter I decided to only use truncated data including winds above 3 m/s and solar radiation above 100 W/m^2. We can find in Appendix A several descriptive statistics on both measures across time, geographical zones, and elevation.

2 Measuring resource complementarity

Using hourly, daily and monthly data, I test the Spearman and Pearson correlations. make a sentence for both measures, explain the differences, I can quote cauz and al I guess.

We distinguish here data at hourly, daily and monthly level. Hourly measures permit to identify very short term variation, and study within a day a complementarity. Daily data identify variation between days (one day cloudy, and windy, with another clear but with low wind). While monthly data attenuate short term variations and identifies seasonal profiles.

I distinguish seasonal profiles since radiations are less likely to be high during winter because of the number of sunny hours and clouds presence. I also compute simultaneous observation in the same station at the same hour using Spearman and Pearson correlations. Finally I explore the frequency at which a resource is above its potential production threshold. I am seeking these results because wind could be high in speed during winter without being high when radiation are low, for instance.

I use the entire dataset from MeteoSwiss from 2020 to 2023 selecting relevant variables as baseline and then I applied a filter to keep only measures above the fixed thresholds (3 m/s for mean wind speed and 100 W/m^2 for mean solar radiation). Thus I use this new sample as a conditional analysis in the sense that, only these measures allow for a potential electricity production. I would have no sense to explore relations between measures with a half of measures not useful to our purpose – explore complementarity between wind and solar radiation for wind turbine and PV installation. Doing this, I filter cases where wind speed is above 3 m/s and solar radiation below 100 W/m^2, meaning that i am not exploring the part where only wind turbines produce (especially during night) and when only PV produce on this subsample. That is why I display results with and without threshold when it is relevant.

An important point to note when threshold are applied. At hourly level I keep only the data above both thresholds, while at daily level I keep the measure at the daily level only if both means are above the thresholds, meaning that during the day, both thresholds are respected but when comes night, the mean of solar radiation drop and is inferior to the threshold dropping the measure for the whole day. Thus, I keep this day for the hourly analysis, but not at the daily one. The same process is applied to the monthly measurements.

3 Results: overall complementarity

My findings are resumed in Table 1 and Table 2. While complementary figures are available in subsection A.1 with the inclusion of seasonal, geographical and elevation controls.

Without thresholds, the Pearson correlation is positive at hourly level and is slightly negative at daily and monthly levels while the Spearman correlation is always positive but not strong. In the filtered subsample, the Pearson and Spearman correlations are

Table 1: With Threshold

Level	Pearson	Spearman
Hourly	-0.1258	-0.0965
Daily	-0.1713	-0.1580
Monthly	-0.2392	-0.1439

Table 2: Without Threshold

Level	Pearson	Spearman
Hourly	0.0925	0.1333
Daily	-0.0459	0.0770
Monthly	-0.0180	0.0936

Table 3: Cauz et al. (2025)

Level	Spearman
Hourly	-0.176
Daily	-0.368
Monthly	-0.736

negative. On the whole dataset, we remain cautious when it comes to complementarity since Spearman correlations are not negatives at all level.

As in the Cauz et al. [2025, Table 1], I display in four zones the different combinations of wind and radiation to enlighten in a different way such a complementarity.

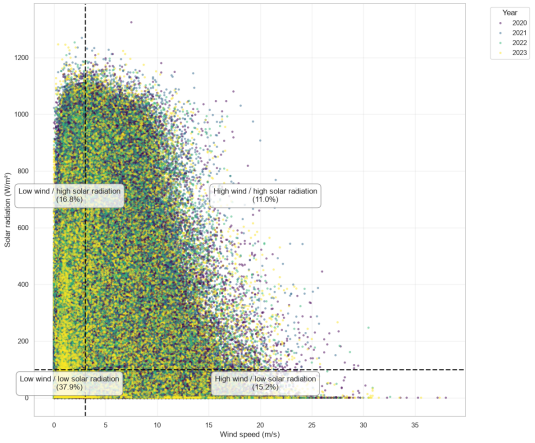

Figure 3: Proportion of wind and radiation conditional on thresholds

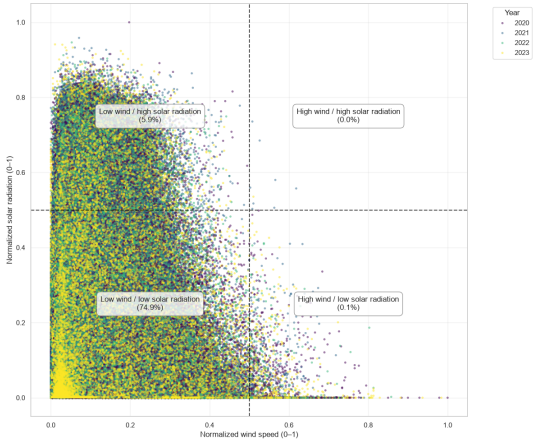

Figure 4: Proportion of wind and radiation without thresholds

Ici rédiger un court paragraphe avec une comparaison avec et sans seuils des différentes proportions.

As said earlier in the introduction, I separated the data across areas, time and elevation. I test correlation between wind speed and solar radiation depending on area, namely: North, South, East and West. In addition, I run tests correlation across seasons and different levels of elevation.

3.1 Results: temporal and seasonal patterns

As noted earlier in Figure 2, there is a seasonality component both for mean wind speed and solar radiation. Whether in the whole sample or in the subsample, we find mean wind speed higher in winter while mean solar radiation is higher during summer and spring compare to other seasons (Table 4 and Table 3.1). Thus, there's a strong seasonal component between the two measures suggesting complementarity between seasons. The combination of wind and solar electricity production has indeed a interest conditionally on season meteorological differences. This suggests – as we will see later – a subsequent

negative correlation between wind and solar measures during cold months (usually December until May, as suggests a previous analysis of mine). This result is present in Table 5 where hourly and daily samples with thresholds display for all seasons slight negatives correlations, especially during autumn and winter.

Table 4: Seasonal wind speed and solar radiation statistics based on daily observations, with and without thresholds.

Sample	Season	Mean wind speed (m/s)	Mean solar radiation (W/m^2)
Without thresholds	Winter	2.90	64.87
Without thresholds	Spring	2.78	201.19
Without thresholds	Summer	2.43	241.03
Without thresholds	Autumn	2.38	111.81
With thresholds	Winter	5.69	132.54
With thresholds	Spring	4.83	226.09
With thresholds	Summer	4.44	257.29
With thresholds	Autumn	4.83	162.41

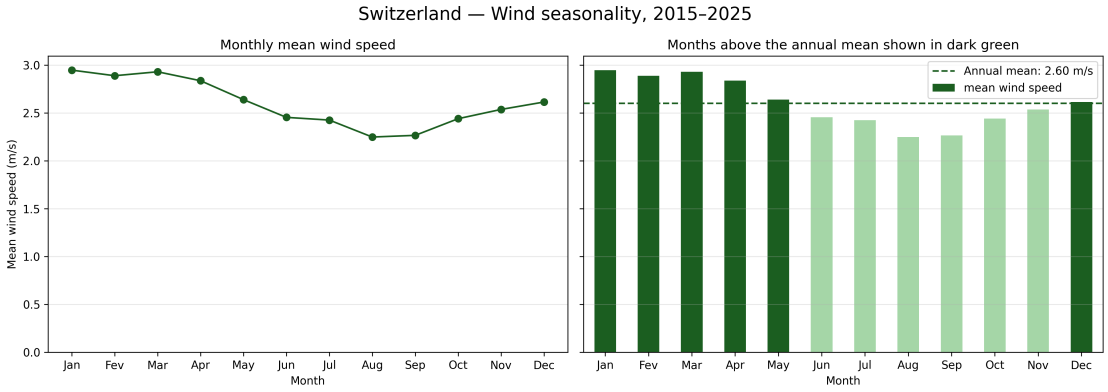

figure to uptade at 2020-2023 levels.

Table 5: Global correlations with thresholds

Scale	Season	N	Pearson	Spearman
Hourly	All	604,689	-0.1258	-0.0965
	Winter	88,338	-0.0588	-0.0498
	Spring	212,728	-0.0717	-0.0465
	Summer	199,963	-0.0501	-0.0267
	Autumn	103,660	-0.1436	-0.1191
Daily	All	32,108	-0.1713	-0.1578
	Winter	3,278	-0.0431	-0.0372
	Spring	12,929	-0.1086	-0.0879
	Summer	10,821	-0.1029	-0.0896
	Autumn	5,080	-0.1756	-0.1815
Monthly	All	1,222	-0.2392	-0.1477
	Winter	108	0.0588	0.1435
	Spring	523	0.0050	0.0896
	Summer	382	-0.1083	-0.0861
	Autumn	209	-0.3089	-0.2406

3.2 Results: altitude and geographical heterogeneity

Altitude matters when it comes to higher speed wind, while – except for presence of clouds – elevation does not matter when it comes to solar radiations. Overall, elevation higher than 1200 meters presents a stronger negative correlation between wind and solar radiation. Figure 5 and Figure 6 provide visual evidences of such a profile, where on average wind speeds increase with elevation.

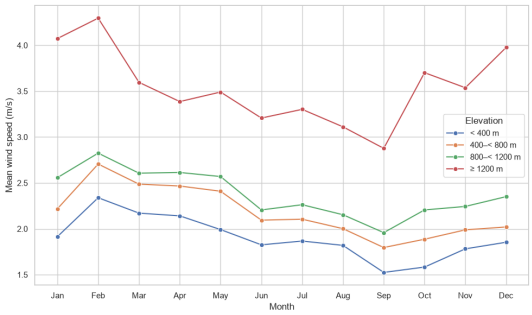

Figure 5: Mean wind speed in whole sample

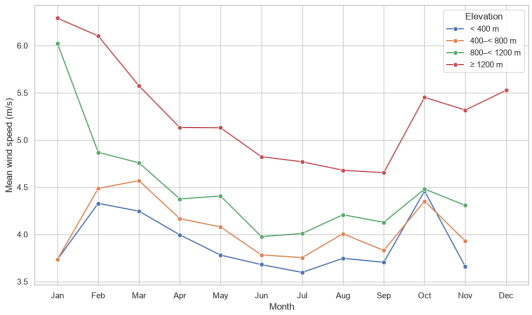

Figure 6: Mean wind speed in the subsample

These stronger winds imply higher negative correlation between wind an solar as

displayed in Table 6. For both Pearson and Spearman correlations, in the whole sample and in the subsample, above 1200 meters, every correlations are negative suggesting wind–solar complementarity.

Table 6: Pearson and Spearman correlations by altitude class

Scale	**Altitude**	**Without Threshold**		**With Threshold**	
		Pearson	Spearman	Pearson	Spearman
Hourly	$<$ 400 m	0.2513	0.2548	-0.0344	-0.0302
	400–$<$ 800 m	0.2016	0.2413	-0.0124	0.0033
	800–$<$ 1200 m	0.1238	0.2118	-0.1497	-0.1193
	$\geq$ 1200 m	-0.0111	-0.0304	-0.1753	-0.1543
Daily	$<$ 400 m	0.0070	0.1187	-0.1458	-0.1356
	400–$<$ 800 m	-0.0206	0.0634	-0.1205	-0.1086
	800–$<$ 1200 m	-0.0421	0.0901	-0.1002	-0.0853
	$\geq$ 1200 m	-0.1323	-0.0174	-0.2009	-0.1939
Monthly	$<$ 400 m	-0.0096	0.0284	-0.1147	-0.1084
	400–$<$ 800 m	0.0253	0.0678	-0.1310	-0.0978
	800–$<$ 1200 m	-0.0538	0.0545	-0.1566	-0.1090
	$\geq$ 1200 m	-0.1248	-0.0082	-0.3190	-0.2412

Elevation depends on the topographic situation of a given country. That is we crossed topographic data with location (North, South, East, West) providing a bigger picture to our analysis. Results suggest a stronger negative correlation in the northern and western part of the country. I display below in Figure 7 and Figure 8 correlations using heatmaps. I appears that in the whole sample that wind–solar negative correlation is higher in +1200 meters northern country part, while in the subsample displays higher negative correlation both between 800m and 1200+m across northern and western country areas.

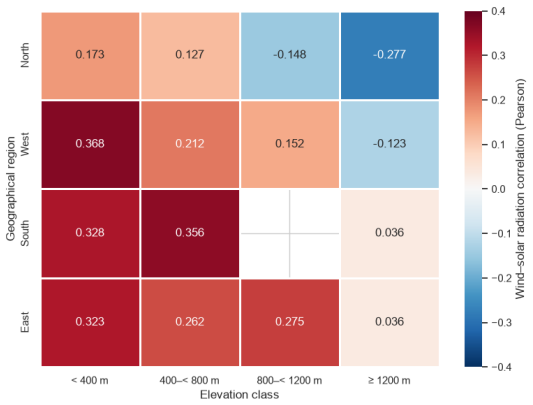

Figure 7: Mean wind speed across months and elevation, with threshold

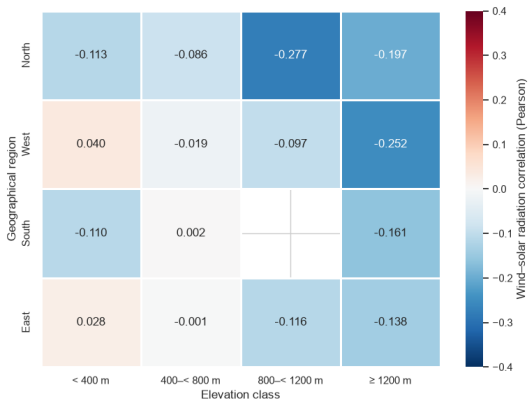

Figure 8: Wind–solar correlation across altitude and location, with threshold

I forgot to say what correlation I used !! (maybe use spearman first, and below between brackets use Pearson's one)

4 Sensitivity analysis and limitations

Overall, results are sensitive to thresholds and the method to compute correlation. Thresholds analysis provide stronger negative correlations but study correlation across meteorological data able to be at the origin of electricity production. Pearson and Spearman permit to distinguish between linear association and a monotonic one without leading at complementarity.

Fixed effect regressions distinguish between variation sources. While stations FE controls for mean invariant differences across station location, temporal FE controls for a same variation across all stations for each observed period. Thus, estimated coefficients describe a conditional relation, not a causal one. Change in sign after the introduction of FE suggests common variation across stations.

In addition, stations group correlations encompass in-group variation invisible at aggregated level. That is why it is important to distinguish stations through altitude and location. Additional checks remain necessary as variation in thresholds; in the final number of observation, especially for monthly measures; in the studied years, *etc.*

Not to forget that we are studying meteorological data complementarity and not electricity productions complementarities. Thus, we need to remain caution when it comes to conclusion in term of complementarity between wind tubrines and PVs installation.

5 Conclusion

This analysis shows an opposition in seasonal profile concerning mean wind speed and solar radiation in Switzerland. Wind is, on average, stronger during winter while solar

radiation are low and vice versa during summer. This suggests a potential seasonal complementarity without short term guarantees.

Results also reveals heterogeneity in elevation and location. Negatives correlations in high places in the northern and western part make room for a potential complementarity – strongly dependent on topographic location.

On the whole sample, correlations remain weak while filtered data display higher negative correlation. Thus, results cannot lead us to a systematic strong and uniform complementarity in Switzerland.

These results favour a precise analysis in term of location to identify most favourable combinations. The interest in this work reside in the conversion of meteorological resources to potential electricity production in order to measure a stable electricity production and its contribution to demand coverage.

A Descriptive statistics

A.1 additional controls on correlations measures

Table 7: Résultats des régressions Vent/Solaire par zone géographique

Échelle	Modèle	Zone	Beta	Erreur Standard	P-value
1. Horaire	Simple	North	7.3522	5.1234	1.5128e-01
		South	14.1971	6.1982	2.1991e-02
		West	6.3674	3.7379	8.8482e-02
		East	10.7395	3.9559	6.6315e-03
	EF Station	North	9.5663	6.1728	1.2120e-01
		South	18.2450	8.1525	2.5225e-02
		West	8.5833	5.1065	9.2789e-02
		East	12.8311	5.3767	1.7013e-02
	EF Temps	North	0.0971	0.3355	7.7229e-01
		South	4.8450	0.9729	6.3661e-07
		West	1.4337	0.4152	5.5366e-04
		East	3.4087	0.6029	1.5727e-08
	EF Station + Temps	North	0.0671	0.4794	8.8876e-01
		South	5.6508	1.1528	9.4895e-07
		West	1.1053	0.5793	5.6380e-02
		East	2.1561	0.6173	4.7762e-04
2. Journalier	Simple	North	-8.9242	1.2305	4.1100e-13
		South	1.9354	2.8243	4.9319e-01
		West	-4.1680	0.8998	3.6215e-06
		East	0.8026	1.4888	5.8983e-01
	EF Station	North	-13.6621	1.8426	1.2257e-13
		South	1.7499	4.6915	7.0915e-01
		West	-9.6085	1.6934	1.3965e-08
		East	-3.7039	2.4412	1.2921e-01
	EF Temps	North	0.2595	0.4084	5.2514e-01
		South	4.7154	1.1160	2.3865e-05
		West	1.1119	0.5175	3.1650e-02
		East	4.0724	0.7777	1.6409e-07
	EF Station + Temps	North	0.1980	0.6658	7.6614e-01
		South	6.2364	1.4114	9.9406e-06
		West	0.1899	0.5582	7.3366e-01
		East	2.2578	0.7780	3.7102e-03
3. Mensuel	Simple	North	-11.0312	3.3376	9.5478e-04
		South	2.8684	4.0465	4.7844e-01
		West	-3.2094	1.7084	6.0347e-02
		East	3.2070	2.2461	1.5339e-01
	EF Station	North	-38.3589	9.8386	9.7715e-05
		South	5.2669	19.6462	7.8864e-01
		West	-24.8229	7.3995	7.9949e-04
		East	-8.6941	9.2917	3.4947e-01
	EF Temps	North	-0.4438	0.6862	5.1778e-01
		South	3.1088	1.4876	3.6679e-02
		West	1.5400	0.9187	9.3727e-02
		East	5.3984	1.0682	4.4601e-07
	EF Station + Temps	North	-2.0719	2.2962	3.6694e-01
		South	5.8456	2.3956	1.4711e-02
		West	-0.6744	1.1654	5.6282e-01
		East	2.5934	1.2090	3.1985e-02

Table 8: Résultats des régressions de la relation Vent/Solaire

Échelle	Modèle	Beta	Erreur Standard	P-value	Obs (N)
1. Horaire	Simple	9.0325	2.3380	1.1187e-04	4,453,593
	EF Station	11.4943	3.0639	1.7575e-04	4,453,593
	EF Temps	2.5051	0.3366	9.8810e-14	4,453,593
	EF Station + Temps	1.9891	0.3948	4.7044e-07	4,453,593
2. Journalier	Simple	-2.2008	0.7790	4.7261e-03	185,736
	EF Station	-6.8185	1.3843	8.4214e-07	185,736
	EF Temps	2.6294	0.4566	8.4636e-09	185,736
	EF Station + Temps	1.7456	0.4989	4.6727e-04	185,736
3. Mensuel	Simple	-0.9140	1.8107	6.1371e-01	6,114
	EF Station	-18.1413	7.2374	1.2217e-02	6,114
	EF Temps	3.0738	0.7121	1.6095e-05	6,114
	EF Station + Temps	1.1248	0.9513	2.3711e-01	6,114

Table 9: decriptive statistics: hourly level, with threshold

	fkl010h0	gre000h0
count	604689.0000	604689.0000
mean	5.1873	459.5605
std	2.2029	257.9068
min	3.0000	100.0000
25%	3.6000	231.0000
50%	4.5000	415.0000
75%	6.1000	665.0000
max	30.5000	1324.0000

Table 10: decriptive statistics: hourly level, without threshold

	fkl010h0	gre000h0
count	5256311.0000	4671553.0000
mean	2.6229	155.2391
std	2.4309	247.0414
min	0.0000	0.0000
25%	1.0000	1.0000
50%	1.8000	6.0000
75%	3.5000	228.0000
max	38.0000	1324.0000

Table 11: Wind and solar radiation correlations with thresholds (Daily)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
0	Daily	Altitude class	400–< 800 m	All periods	7,708	-0.1205	-0.1086
1	Daily	Altitude class	400–< 800 m	Autumn	835	-0.2357	-0.2180
2	Daily	Altitude class	400–< 800 m	Spring	3,859	-0.0703	-0.0581
3	Daily	Altitude class	400–< 800 m	Summer	2,447	-0.0316	-0.0377
4	Daily	Altitude class	400–< 800 m	Winter	567	-0.0314	-0.0276
5	Daily	Altitude class	800–< 1200 m	All periods	3,255	-0.1002	-0.0853
6	Daily	Altitude class	800–< 1200 m	Autumn	502	-0.1021	-0.1163
7	Daily	Altitude class	800–< 1200 m	Spring	1,392	-0.0451	-0.0292
8	Daily	Altitude class	800–< 1200 m	Summer	1,106	0.0070	0.0032
9	Daily	Altitude class	800–< 1200 m	Winter	255	-0.1325	-0.1288
10	Daily	Altitude class	< 400 m	All periods	1,179	-0.1458	-0.1356
11	Daily	Altitude class	< 400 m	Autumn	110	-0.2601	-0.3180
12	Daily	Altitude class	< 400 m	Spring	576	-0.0698	-0.0551
13	Daily	Altitude class	< 400 m	Summer	359	0.0676	0.0277
14	Daily	Altitude class	< 400 m	Winter	134	-0.1480	-0.0658
15	Daily	Altitude class	≥ 1200 m	All periods	19,963	-0.2009	-0.1939
16	Daily	Altitude class	≥ 1200 m	Autumn	3,633	-0.1934	-0.1940
17	Daily	Altitude class	≥ 1200 m	Spring	7,103	-0.1790	-0.1653
18	Daily	Altitude class	≥ 1200 m	Summer	6,910	-0.1271	-0.1239
19	Daily	Altitude class	≥ 1200 m	Winter	2,317	-0.0860	-0.0766
20	Daily	Geographic zone	East	All periods	12,565	-0.2072	-0.1982
21	Daily	Geographic zone	East	Autumn	1,989	-0.1891	-0.2031
22	Daily	Geographic zone	East	Spring	4,770	-0.1249	-0.1165
23	Daily	Geographic zone	East	Summer	4,485	-0.1353	-0.1096
24	Daily	Geographic zone	East	Winter	1,321	-0.0746	-0.0835
25	Daily	Geographic zone	North	All periods	4,768	-0.1737	-0.1623
26	Daily	Geographic zone	North	Autumn	719	-0.1973	-0.2060
27	Daily	Geographic zone	North	Spring	2,154	-0.1305	-0.1023
28	Daily	Geographic zone	North	Summer	1,450	-0.1101	-0.1073
29	Daily	Geographic zone	North	Winter	445	-0.0079	-0.0355
30	Daily	Geographic zone	South	All periods	5,651	-0.1936	-0.1639
31	Daily	Geographic zone	South	Autumn	961	-0.1720	-0.1588
32	Daily	Geographic zone	South	Spring	2,088	-0.1644	-0.1165
33	Daily	Geographic zone	South	Summer	1,917	-0.1085	-0.0820
34	Daily	Geographic zone	South	Winter	685	-0.0673	-0.0278
35	Daily	Geographic zone	West	All periods	9,121	-0.1414	-0.1234
36	Daily	Geographic zone	West	Autumn	1,411	-0.1648	-0.1594
37	Daily	Geographic zone	West	Spring	3,918	-0.0928	-0.0679
38	Daily	Geographic zone	West	Summer	2,970	-0.0883	-0.0869
39	Daily	Geographic zone	West	Winter	822	-0.0177	0.0053
40	Daily	Overall	All stations	All periods	32,105	-0.1719	-0.1580
41	Daily	Overall	All stations	Autumn	5,080	-0.1748	-0.1806
42	Daily	Overall	All stations	Spring	12,930	-0.1103	-0.0887
43	Daily	Overall	All stations	Summer	10,822	-0.1028	-0.0895
44	Daily	Overall	All stations	Winter	3,273	-0.0437	-0.0379

Table 12: Wind and solar radiation correlations with thresholds (Hourly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
45	Hourly	Altitude class	400–< 800 m	All periods	205,930	-0.0124	0.0033
46	Hourly	Altitude class	400–< 800 m	Autumn	32,079	-0.0659	-0.0440
47	Hourly	Altitude class	400–< 800 m	Spring	80,265	0.0152	0.0258
48	Hourly	Altitude class	400–< 800 m	Summer	68,652	0.0808	0.0813
49	Hourly	Altitude class	400–< 800 m	Winter	24,934	-0.0264	-0.0306
50	Hourly	Altitude class	800–< 1200 m	All periods	75,620	-0.1497	-0.1193
51	Hourly	Altitude class	800–< 1200 m	Autumn	12,420	-0.1399	-0.1297
52	Hourly	Altitude class	800–< 1200 m	Spring	27,770	-0.0849	-0.0570
53	Hourly	Altitude class	800–< 1200 m	Summer	26,600	-0.0579	-0.0291
54	Hourly	Altitude class	800–< 1200 m	Winter	8,830	-0.0649	-0.0724
55	Hourly	Altitude class	< 400 m	All periods	43,634	-0.0344	-0.0302
56	Hourly	Altitude class	< 400 m	Autumn	6,002	-0.0785	-0.0814
57	Hourly	Altitude class	< 400 m	Spring	17,473	0.0156	0.0230
58	Hourly	Altitude class	< 400 m	Summer	14,348	0.0631	0.0644
59	Hourly	Altitude class	< 400 m	Winter	5,811	0.0401	0.0249
60	Hourly	Altitude class	≥ 1200 m	All periods	279,505	-0.1753	-0.1543
61	Hourly	Altitude class	≥ 1200 m	Autumn	53,159	-0.1867	-0.1683
62	Hourly	Altitude class	≥ 1200 m	Spring	87,220	-0.1372	-0.1168
63	Hourly	Altitude class	≥ 1200 m	Summer	90,363	-0.1000	-0.0823
64	Hourly	Altitude class	≥ 1200 m	Winter	48,763	-0.1156	-0.1073
65	Hourly	Geographic zone	East	All periods	210,337	-0.1319	-0.1054
66	Hourly	Geographic zone	East	Autumn	36,282	-0.1554	-0.1319
67	Hourly	Geographic zone	East	Spring	71,700	-0.0766	-0.0633
68	Hourly	Geographic zone	East	Summer	71,494	-0.0639	-0.0448
69	Hourly	Geographic zone	East	Winter	30,861	-0.0621	-0.0508
70	Hourly	Geographic zone	North	All periods	101,983	-0.1505	-0.1320
71	Hourly	Geographic zone	North	Autumn	16,874	-0.1392	-0.1163
72	Hourly	Geographic zone	North	Spring	38,268	-0.0925	-0.0674
73	Hourly	Geographic zone	North	Summer	30,030	-0.0583	-0.0498
74	Hourly	Geographic zone	North	Winter	16,811	-0.0678	-0.0549
75	Hourly	Geographic zone	South	All periods	93,126	-0.1386	-0.1136
76	Hourly	Geographic zone	South	Autumn	15,789	-0.1722	-0.1486
77	Hourly	Geographic zone	South	Spring	31,665	-0.0889	-0.0713
78	Hourly	Geographic zone	South	Summer	31,960	-0.0594	-0.0505
79	Hourly	Geographic zone	South	Winter	13,712	-0.0867	-0.0787
80	Hourly	Geographic zone	West	All periods	199,243	-0.1389	-0.0953
81	Hourly	Geographic zone	West	Autumn	34,715	-0.1576	-0.1281
82	Hourly	Geographic zone	West	Spring	71,095	-0.0803	-0.0365
83	Hourly	Geographic zone	West	Summer	66,479	-0.0600	-0.0155
84	Hourly	Geographic zone	West	Winter	26,954	-0.0888	-0.0768
85	Hourly	Overall	All stations	All periods	604,689	-0.1258	-0.0965
86	Hourly	Overall	All stations	Autumn	103,660	-0.1436	-0.1191
87	Hourly	Overall	All stations	Spring	212,728	-0.0717	-0.0465
88	Hourly	Overall	All stations	Summer	199,963	-0.0501	-0.0267
89	Hourly	Overall	All stations	Winter	88,338	-0.0588	-0.0498

Table 13: Wind and solar radiation correlations with thresholds (Monthly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
90	Monthly	Altitude class	400–< 800 m	All periods	222	-0.1310	-0.0978
91	Monthly	Altitude class	400–< 800 m	Autumn	15	-0.2506	-0.2286
92	Monthly	Altitude class	400–< 800 m	Spring	151	-0.1302	-0.0924
93	Monthly	Altitude class	400–< 800 m	Summer	48	-0.0334	-0.0098
94	Monthly	Altitude class	400–< 800 m	Winter	8	0.0286	0.1190
95	Monthly	Altitude class	800–< 1200 m	All periods	109	-0.1566	-0.1090
96	Monthly	Altitude class	800–< 1200 m	Autumn	14	-0.4384	-0.4725
97	Monthly	Altitude class	800–< 1200 m	Spring	60	-0.2099	-0.1789
98	Monthly	Altitude class	800–< 1200 m	Summer	27	0.1231	0.0714
99	Monthly	Altitude class	800–< 1200 m	Winter	8	-0.2253	-0.3333
100	Monthly	Altitude class	< 400 m	All periods	18	-0.1147	-0.1084
101	Monthly	Altitude class	< 400 m	Spring	8	-0.0084	-0.0238
102	Monthly	Altitude class	< 400 m	Summer	8	0.5205	0.6667
103	Monthly	Altitude class	< 400 m	Winter	2	1.0000	1.0000
104	Monthly	Altitude class	≥ 1200 m	All periods	871	-0.3190	-0.2412
105	Monthly	Altitude class	≥ 1200 m	Autumn	180	-0.3641	-0.3049
106	Monthly	Altitude class	≥ 1200 m	Spring	304	-0.2298	-0.1239
107	Monthly	Altitude class	≥ 1200 m	Summer	299	-0.1208	-0.1029
108	Monthly	Altitude class	≥ 1200 m	Winter	88	-0.2193	-0.1737
109	Monthly	Geographic zone	East	All periods	507	-0.2742	-0.1580
110	Monthly	Geographic zone	East	Autumn	93	-0.3741	-0.2548
111	Monthly	Geographic zone	East	Spring	186	0.0640	0.1642
112	Monthly	Geographic zone	East	Summer	176	-0.1811	-0.1328
113	Monthly	Geographic zone	East	Winter	52	0.2718	0.2645
114	Monthly	Geographic zone	North	All periods	174	-0.3506	-0.3200
115	Monthly	Geographic zone	North	Autumn	30	-0.5946	-0.5519
116	Monthly	Geographic zone	North	Spring	83	-0.2523	-0.2568
117	Monthly	Geographic zone	North	Summer	57	0.0926	0.0432
118	Monthly	Geographic zone	North	Winter	4	0.1416	0.2000
119	Monthly	Geographic zone	South	All periods	233	-0.2830	-0.2061
120	Monthly	Geographic zone	South	Autumn	43	-0.2217	-0.1924
121	Monthly	Geographic zone	South	Spring	87	0.0082	0.0406
122	Monthly	Geographic zone	South	Summer	78	-0.1789	-0.1781
123	Monthly	Geographic zone	South	Winter	25	-0.1811	-0.1808
124	Monthly	Geographic zone	West	All periods	306	-0.1907	-0.1069
125	Monthly	Geographic zone	West	Autumn	43	-0.3438	-0.3143
126	Monthly	Geographic zone	West	Spring	167	-0.0438	0.0226
127	Monthly	Geographic zone	West	Summer	71	-0.2301	-0.2170
128	Monthly	Geographic zone	West	Winter	25	0.1205	0.2754
129	Monthly	Overall	All stations	All periods	1,220	-0.2336	-0.1439
130	Monthly	Overall	All stations	Autumn	209	-0.3102	-0.2420
131	Monthly	Overall	All stations	Spring	523	0.0054	0.0903
132	Monthly	Overall	All stations	Summer	382	-0.1098	-0.0855
133	Monthly	Overall	All stations	Winter	106	0.0946	0.1785

Table 14: Wind and solar radiation correlations without thresholds (Daily)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
0	Daily	Altitude class	400–< 800 m	All periods	95,339	-0.0206	0.0634
1	Daily	Altitude class	400–< 800 m	Autumn	23,749	-0.1269	-0.0822
2	Daily	Altitude class	400–< 800 m	Spring	24,000	-0.0529	-0.0171
3	Daily	Altitude class	400–< 800 m	Summer	24,011	0.0825	0.0889
4	Daily	Altitude class	400–< 800 m	Winter	23,579	-0.0683	-0.0637
5	Daily	Altitude class	800–< 1200 m	All periods	36,506	-0.0421	0.0901
6	Daily	Altitude class	800–< 1200 m	Autumn	9,084	-0.1233	-0.0314
7	Daily	Altitude class	800–< 1200 m	Spring	9,197	-0.0246	0.0747
8	Daily	Altitude class	800–< 1200 m	Summer	9,200	0.1515	0.2432
9	Daily	Altitude class	800–< 1200 m	Winter	9,025	-0.1232	-0.1018
10	Daily	Altitude class	< 400 m	All periods	24,837	0.0070	0.1187
11	Daily	Altitude class	< 400 m	Autumn	6,188	-0.1020	-0.0336
12	Daily	Altitude class	< 400 m	Spring	6,256	0.0010	0.0530
13	Daily	Altitude class	< 400 m	Summer	6,256	0.0175	0.0273
14	Daily	Altitude class	< 400 m	Winter	6,137	-0.0018	0.0342
15	Daily	Altitude class	≥ 1200 m	All periods	72,972	-0.1323	-0.0174
16	Daily	Altitude class	≥ 1200 m	Autumn	18,145	-0.1784	-0.0788
17	Daily	Altitude class	≥ 1200 m	Spring	18,397	-0.0830	0.0285
18	Daily	Altitude class	≥ 1200 m	Summer	18,382	-0.0444	0.0312
19	Daily	Altitude class	≥ 1200 m	Winter	18,048	-0.0474	0.0190
20	Daily	Geographic zone	East	All periods	67,153	0.0187	0.1780
21	Daily	Geographic zone	East	Autumn	16,721	-0.0361	0.0922
22	Daily	Geographic zone	East	Spring	16,916	0.0805	0.1761
23	Daily	Geographic zone	East	Summer	16,910	0.0776	0.1730
24	Daily	Geographic zone	East	Winter	16,606	0.1131	0.1740
25	Daily	Geographic zone	North	All periods	54,440	-0.1458	-0.0702
26	Daily	Geographic zone	North	Autumn	13,559	-0.1703	-0.1507
27	Daily	Geographic zone	North	Spring	13,705	-0.1458	-0.1099
28	Daily	Geographic zone	North	Summer	13,707	-0.0728	-0.0292
29	Daily	Geographic zone	North	Winter	13,469	-0.0491	-0.0393
30	Daily	Geographic zone	South	All periods	32,137	0.0055	0.1758
31	Daily	Geographic zone	South	Autumn	8,006	-0.0612	0.0811
32	Daily	Geographic zone	South	Spring	8,093	0.0746	0.1905
33	Daily	Geographic zone	South	Summer	8,096	0.0195	0.0955
34	Daily	Geographic zone	South	Winter	7,942	0.0813	0.1840
35	Daily	Geographic zone	West	All periods	75,924	-0.0867	0.0239
36	Daily	Geographic zone	West	Autumn	18,880	-0.1419	-0.0846
37	Daily	Geographic zone	West	Spring	19,136	-0.0880	-0.0181
38	Daily	Geographic zone	West	Summer	19,136	0.0017	0.0839
39	Daily	Geographic zone	West	Winter	18,772	-0.0505	-0.0586
40	Daily	Overall	All stations	All periods	229,654	-0.0461	0.0770
41	Daily	Overall	All stations	Autumn	57,166	-0.0936	-0.0136
42	Daily	Overall	All stations	Spring	57,850	-0.0170	0.0574
43	Daily	Overall	All stations	Summer	57,849	0.0150	0.0851
44	Daily	Overall	All stations	Winter	56,789	0.0293	0.0536

Table 15: Wind and solar radiation correlations without thresholds (Hourly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
45	Hourly	Altitude class	400–< 800 m	All periods	2,288,018	0.2016	0.2413
46	Hourly	Altitude class	400–< 800 m	Autumn	569,966	0.1443	0.1908
47	Hourly	Altitude class	400–< 800 m	Spring	575,927	0.2522	0.3072
48	Hourly	Altitude class	400–< 800 m	Summer	576,239	0.3423	0.3778
49	Hourly	Altitude class	400–< 800 m	Winter	565,886	0.0553	0.0701
50	Hourly	Altitude class	800–< 1200 m	All periods	876,079	0.1238	0.2118
51	Hourly	Altitude class	800–< 1200 m	Autumn	217,982	0.0793	0.1650
52	Hourly	Altitude class	800–< 1200 m	Spring	220,697	0.1805	0.2786
53	Hourly	Altitude class	800–< 1200 m	Summer	220,800	0.2780	0.3493
54	Hourly	Altitude class	800–< 1200 m	Winter	216,600	-0.0114	0.0450
55	Hourly	Altitude class	< 400 m	All periods	596,088	0.2513	0.2548
56	Hourly	Altitude class	< 400 m	Autumn	148,512	0.1787	0.1682
57	Hourly	Altitude class	< 400 m	Spring	150,144	0.3262	0.3466
58	Hourly	Altitude class	< 400 m	Summer	150,144	0.3741	0.3888
59	Hourly	Altitude class	< 400 m	Winter	147,288	0.1095	0.0852
60	Hourly	Altitude class	≥ 1200 m	All periods	1,751,100	-0.0111	-0.0304
61	Hourly	Altitude class	≥ 1200 m	Autumn	435,347	-0.0259	-0.0704
62	Hourly	Altitude class	≥ 1200 m	Spring	441,502	0.0186	0.0236
63	Hourly	Altitude class	≥ 1200 m	Summer	441,121	0.0699	0.0836
64	Hourly	Altitude class	≥ 1200 m	Winter	433,130	-0.0231	-0.1255
65	Hourly	Geographic zone	East	All periods	1,611,502	0.1183	0.1412
66	Hourly	Geographic zone	East	Autumn	401,254	0.0728	0.0737
67	Hourly	Geographic zone	East	Spring	405,912	0.1686	0.2177
68	Hourly	Geographic zone	East	Summer	405,793	0.2207	0.2671
69	Hourly	Geographic zone	East	Winter	398,543	0.0504	-0.0224
70	Hourly	Geographic zone	North	All periods	1,306,472	0.0472	0.1079
71	Hourly	Geographic zone	North	Autumn	325,416	0.0249	0.0644
72	Hourly	Geographic zone	North	Spring	328,888	0.1026	0.1679
73	Hourly	Geographic zone	North	Summer	328,943	0.1590	0.2346
74	Hourly	Geographic zone	North	Winter	323,225	0.0199	0.0055
75	Hourly	Geographic zone	South	All periods	771,213	0.0949	0.0785
76	Hourly	Geographic zone	South	Autumn	192,095	0.0514	0.0209
77	Hourly	Geographic zone	South	Spring	194,206	0.1446	0.1498
78	Hourly	Geographic zone	South	Summer	194,304	0.1762	0.1799
79	Hourly	Geographic zone	South	Winter	190,608	0.0273	-0.0711
80	Hourly	Geographic zone	West	All periods	1,822,098	0.0879	0.1688
81	Hourly	Geographic zone	West	Autumn	453,042	0.0547	0.1346
82	Hourly	Geographic zone	West	Spring	459,264	0.1364	0.2255
83	Hourly	Geographic zone	West	Summer	459,264	0.2134	0.3109
84	Hourly	Geographic zone	West	Winter	450,528	0.0110	0.0130
85	Hourly	Overall	All stations	All periods	5,511,285	0.0925	0.1333
86	Hourly	Overall	All stations	Autumn	1,371,807	0.0564	0.0828
87	Hourly	Overall	All stations	Spring	1,388,270	0.1429	0.1999
88	Hourly	Overall	All stations	Summer	1,388,304	0.1970	0.2580
89	Hourly	Overall	All stations	Winter	1,362,904	0.0304	-0.0129

Table 16: Wind and solar radiation correlations without thresholds (Monthly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
90	Monthly	Altitude class	400–< 800 m	All periods	3,133	0.0253	0.0678
91	Monthly	Altitude class	400–< 800 m	Autumn	783	-0.1385	-0.1090
92	Monthly	Altitude class	400–< 800 m	Spring	783	-0.0249	-0.0202
93	Monthly	Altitude class	400–< 800 m	Summer	783	0.1675	0.1748
94	Monthly	Altitude class	400–< 800 m	Winter	784	0.2597	0.2402
95	Monthly	Altitude class	800–< 1200 m	All periods	1,200	-0.0538	0.0545
96	Monthly	Altitude class	800–< 1200 m	Autumn	300	-0.1466	-0.0757
97	Monthly	Altitude class	800–< 1200 m	Spring	300	-0.0786	-0.0230
98	Monthly	Altitude class	800–< 1200 m	Summer	300	0.2496	0.2690
99	Monthly	Altitude class	800–< 1200 m	Winter	300	0.0339	0.0298
100	Monthly	Altitude class	< 400 m	All periods	816	-0.0096	0.0284
101	Monthly	Altitude class	< 400 m	Autumn	204	-0.2192	-0.1653
102	Monthly	Altitude class	< 400 m	Spring	204	-0.2298	-0.2428
103	Monthly	Altitude class	< 400 m	Summer	204	-0.0840	-0.0396
104	Monthly	Altitude class	< 400 m	Winter	204	0.1036	0.0811
105	Monthly	Altitude class	≥ 1200 m	All periods	2,400	-0.1248	-0.0082
106	Monthly	Altitude class	≥ 1200 m	Autumn	600	-0.1392	-0.0276
107	Monthly	Altitude class	≥ 1200 m	Spring	600	0.0426	0.1400
108	Monthly	Altitude class	≥ 1200 m	Summer	600	-0.0154	0.0235
109	Monthly	Altitude class	≥ 1200 m	Winter	600	0.1141	0.1762
110	Monthly	Geographic zone	East	All periods	2,208	0.0719	0.2191
111	Monthly	Geographic zone	East	Autumn	552	0.0631	0.1578
112	Monthly	Geographic zone	East	Spring	552	0.3330	0.3609
113	Monthly	Geographic zone	East	Summer	552	0.1908	0.2449
114	Monthly	Geographic zone	East	Winter	552	0.3199	0.3703
115	Monthly	Geographic zone	North	All periods	1,789	-0.1573	-0.0905
116	Monthly	Geographic zone	North	Autumn	447	-0.1647	-0.1688
117	Monthly	Geographic zone	North	Spring	447	-0.1819	-0.1892
118	Monthly	Geographic zone	North	Summer	447	-0.0434	0.0334
119	Monthly	Geographic zone	North	Winter	448	0.2331	0.2645
120	Monthly	Geographic zone	South	All periods	1,056	0.0290	0.1685
121	Monthly	Geographic zone	South	Autumn	264	-0.0229	0.0939
122	Monthly	Geographic zone	South	Spring	264	0.2211	0.2401
123	Monthly	Geographic zone	South	Summer	264	0.0301	-0.0119
124	Monthly	Geographic zone	South	Winter	264	0.2106	0.2531
125	Monthly	Geographic zone	West	All periods	2,496	-0.0668	0.0253
126	Monthly	Geographic zone	West	Autumn	624	-0.0997	-0.0754
127	Monthly	Geographic zone	West	Spring	624	-0.0461	-0.0211
128	Monthly	Geographic zone	West	Summer	624	-0.0200	0.1017
129	Monthly	Geographic zone	West	Winter	624	0.1938	0.2280
130	Monthly	Overall	All stations	All periods	7,549	-0.0184	0.0936
131	Monthly	Overall	All stations	Autumn	1,887	-0.0394	0.0189
132	Monthly	Overall	All stations	Spring	1,887	0.1041	0.1215
133	Monthly	Overall	All stations	Summer	1,887	0.0409	0.0962
134	Monthly	Overall	All stations	Winter	1,888	0.2493	0.2664

References

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